Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports: A Synthetic Study

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Overview

Six parts of the presentation

01 Introduction

The rescue-report problem

02 Related work

What earlier studies provide

03 Dataset

How we create the test reports

04 Method

Grouping, ranking and rescue trips

05 Results

What the tests show

06 Conclusion & Future work

Main lessons and next steps

Introduction: The rescue-report problem

From uncertain observations to feasible rescue decisions



Before sending a team

01 Identify

Which reports belong to the same incident?

02 Prioritize

Which incident should come first?

03 Schedule

Which trips are feasible with available crews?

Introduction: Why grouping matters

Three illustrative failures change where a team would be sent



Repeated observations

Risk: repeated trips



Distinct nearby incidents

Risk: a missed need



An invented destination

Risk: a false destination

Related work: The gap between steps

Connect information extraction, consolidation and resource allocation

Research strand	What it provides	What still needs testing
Crisis sensing [6–8]	Crisis posts and flood imagery	Which physical incident?
Community detection [1]	Groups of related reports	A genuine rescue destination?
Humanitarian logistics [5]	Allocation and routing trade-offs	Consequences of upstream errors?

Our question: do clustering and scoring safeguards improve dispatch?

Tested in three steps: grouping (RQ1) → ranking (RQ2) → dispatch (RQ3)

Dataset: How we create the reports

A synthetic dataset with controlled observations and known evaluation targets

1 Set the scenario

EMSR848 · OSM · WorldPop (clustering tests)

2 Simulate incidents

16 hidden incidents per run

3 Generate reports

Location · time · evidence

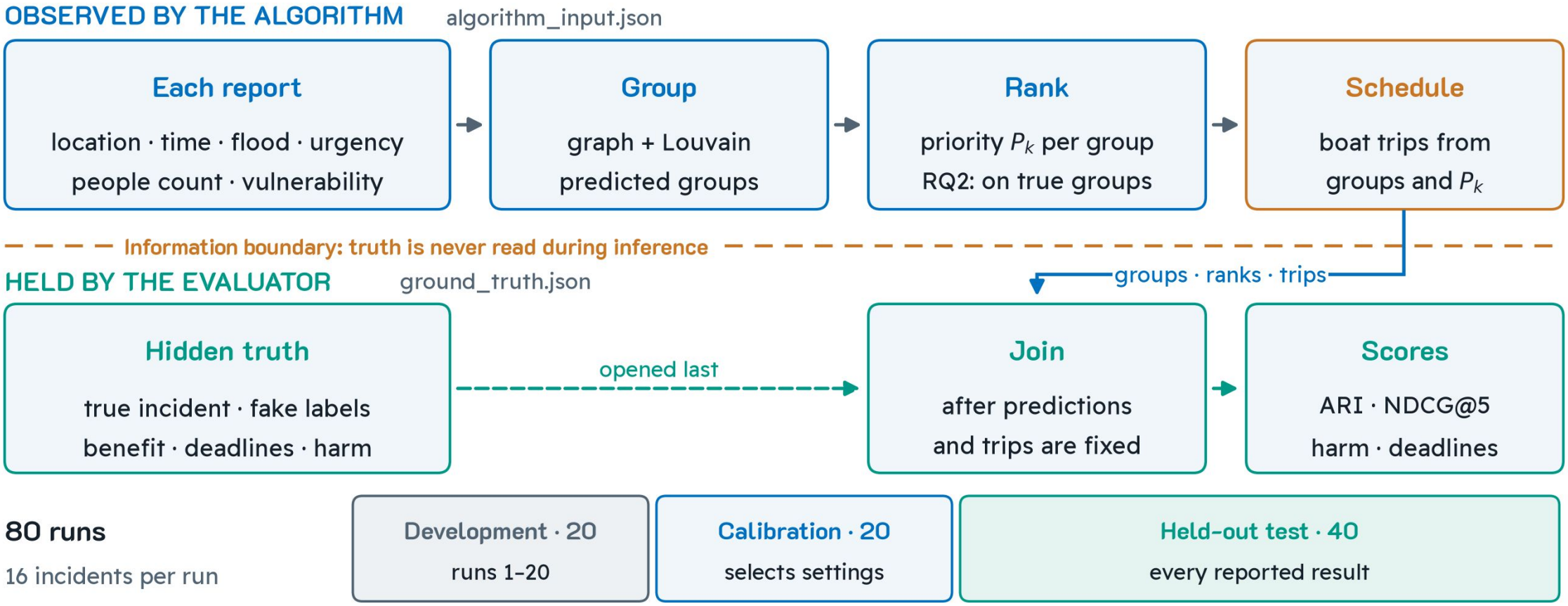
4 Add controlled noise

Copies and false reports



Dataset: Inputs and evaluation truth

The algorithm cannot consult incident identity or simulated outcomes



Method: From reports to incident groups

Two ways to combine distance, time and evidence before community detection

Reports
location · time
flood, urgency

→

Pair similarity
G distance · T time
C evidence

→

Edge weight
Product
or Additive

→

Keep strong links
quantile threshold
then top-k neighbours

→

Louvain
communities =
predicted groups

Product — Distance gates time and evidence, cf. [2]

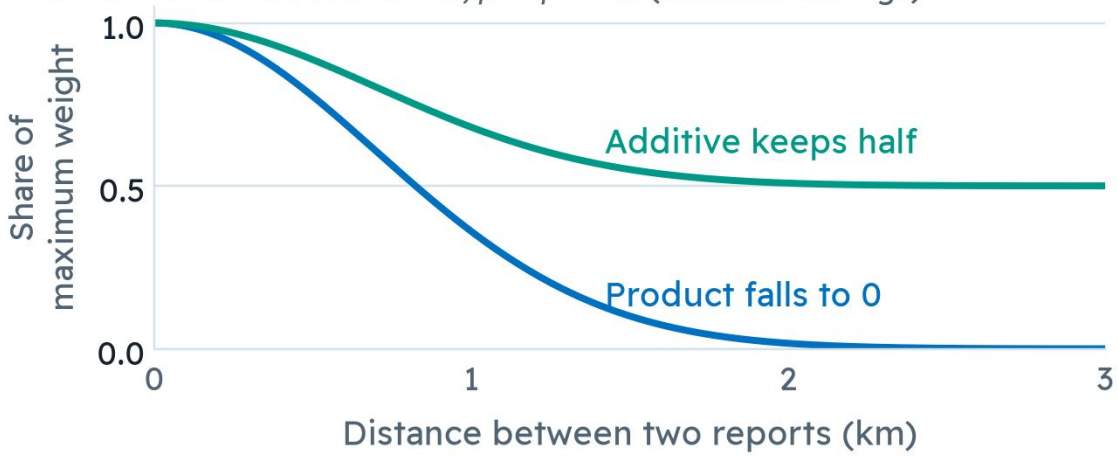
$$w_{ij}^{\times} = G_{ij}\,(\beta\,T_{ij} + \gamma\,C_{ij})$$

Additive — Each signal adds on its own

$$w_{ij}^{+} = \alpha\,G_{ij} + \beta\,T_{ij} + \gamma\,C_{ij}$$

Two reports with the same time and evidence

$T = C = 1 \cdot \sigma = 700$ m · $\alpha = 1$, $\beta = \gamma = 0.5$ (selected settings)



Method: From evidence to rescue trips

Evidence families, a bounded score and a scheduler solve different problems

Predicted group
reports in one cluster

→ **Evidence families**
exact copies and near copies together

→ **One value per family**
family maximum × report confidence Q_i

→ **Priority P_k**
bounded score for the group

→ **Scheduler**
travel and service time available boats

Exact-copy guarantee

An exact copy joins an existing family, so the family maximum and the score stay the same.

Holds only when families and score inputs are fixed.

Bounded priority $0 \le P_k \le \mu$

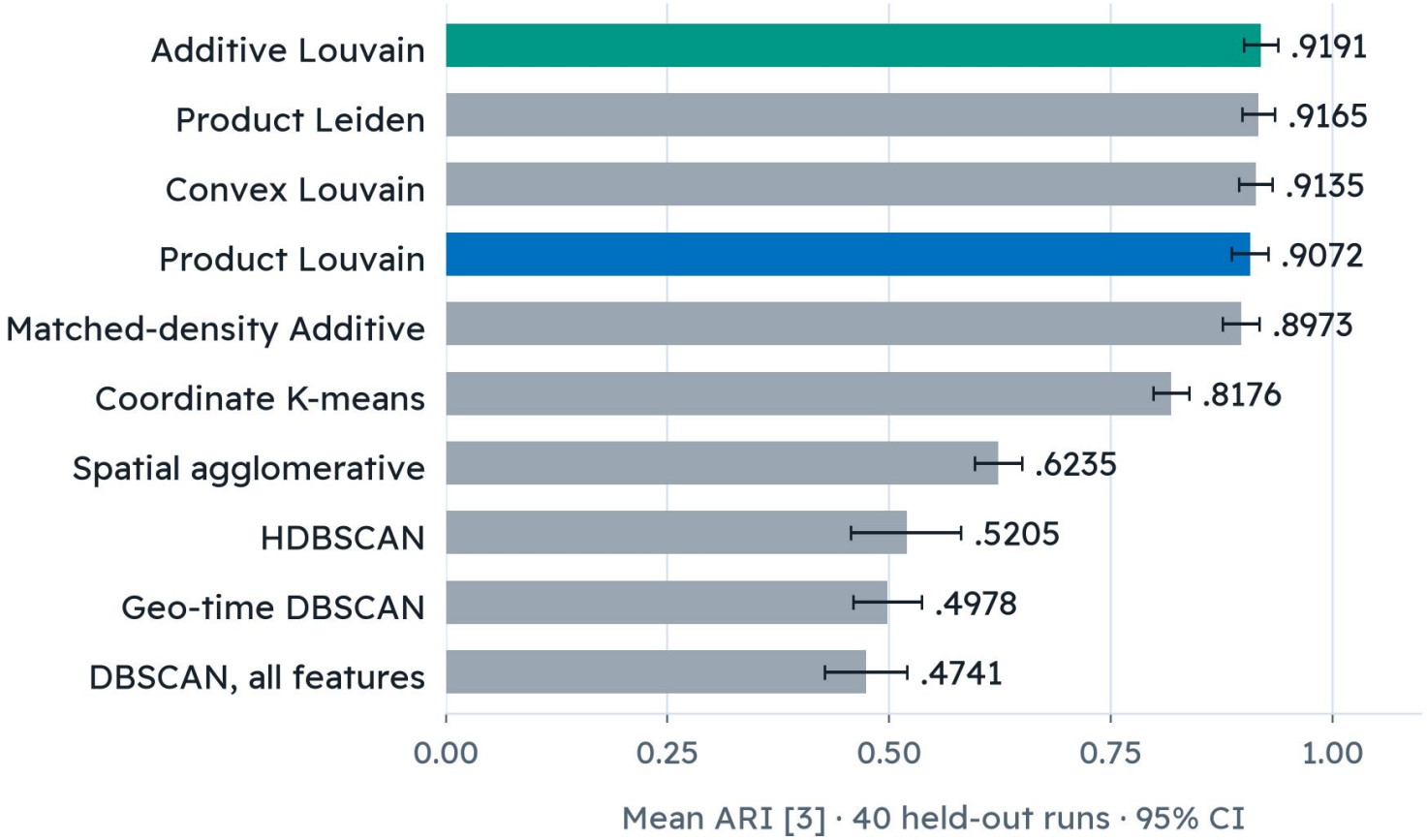
$$P_k = \underbrace{\left[1 + (\mu - 1)\tanh(\bar{V}_k/s)\right]}_{\text{vulnerability boost}} \underbrace{(\omega_E \bar{E}_k + \omega_F \bar{F}_k + \omega_N \bar{N}_k)}_{\text{urgency, flood, people}}$$

vulnerability boost between 1 and $\mu = 2$

urgency, flood, people each in [0, 1], weights sum to 1

Results: Clustering accuracy

Adjusted Rand Index of independently tuned methods · 40 held-out runs



Highest mean: Additive

ARI .9191 vs .9072
for Product

No advantage established

Paired Wilcoxon,
Holm p = .2560

Graph methods lead

Other baselines:
ARI .47 to .82

Results: Clustering under stress

Seven fixed conditions on the 40 held-out runs · settings not retuned

	Control	Location ±100 m	Location ±300 m	Time ±15 min	Time ±60 min	Copies 2×	Copies 5×
Additive Louvain	.92	.90	.76	.88	.76	.93	.30
Product Louvain	.91	.90	.78	.87	.74	.93	.30
Product Leiden	.92	.90	.79	.89	.76	.93	.30
Matched-density Additive	.90	.88	.80	.85	.71	.92	.29
Geo-time DBSCAN	.50	.47	.37	.49	.44	.50	.50

Mean ARI over 40 held-out runs · original reports scored in every column

5× exact copies

All graph variants: ARI ≈ .30

(Product ΔARI = −.6108)

Location and time noise

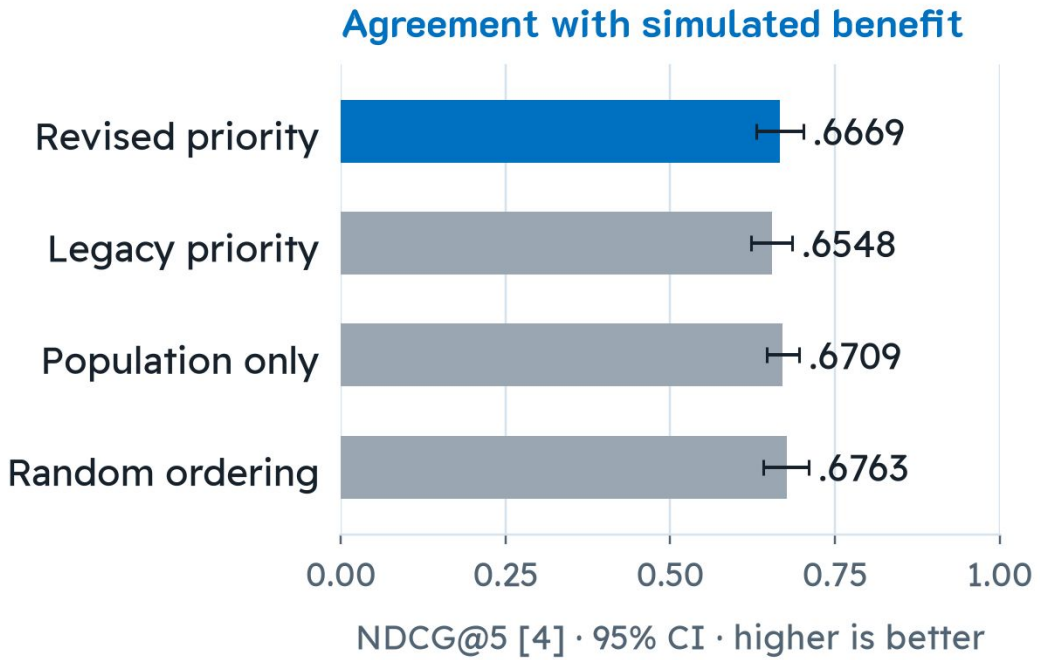
Every tested level lowers mean ARI for all methods

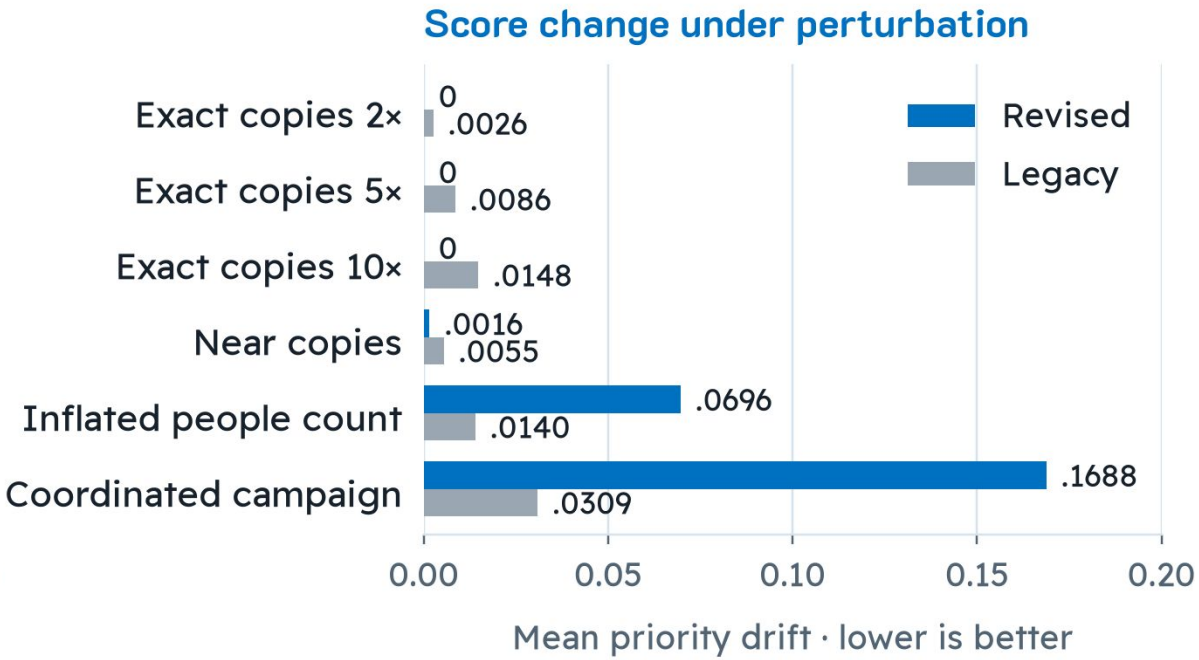
False destinations

16.49% rate with Product: fake-only groups still form

Results: Ranking

Copy invariance and benefit alignment answer different questions





No alignment advantage

+.0121 vs legacy, not significant;
below population-only and random

Stable under exact copies

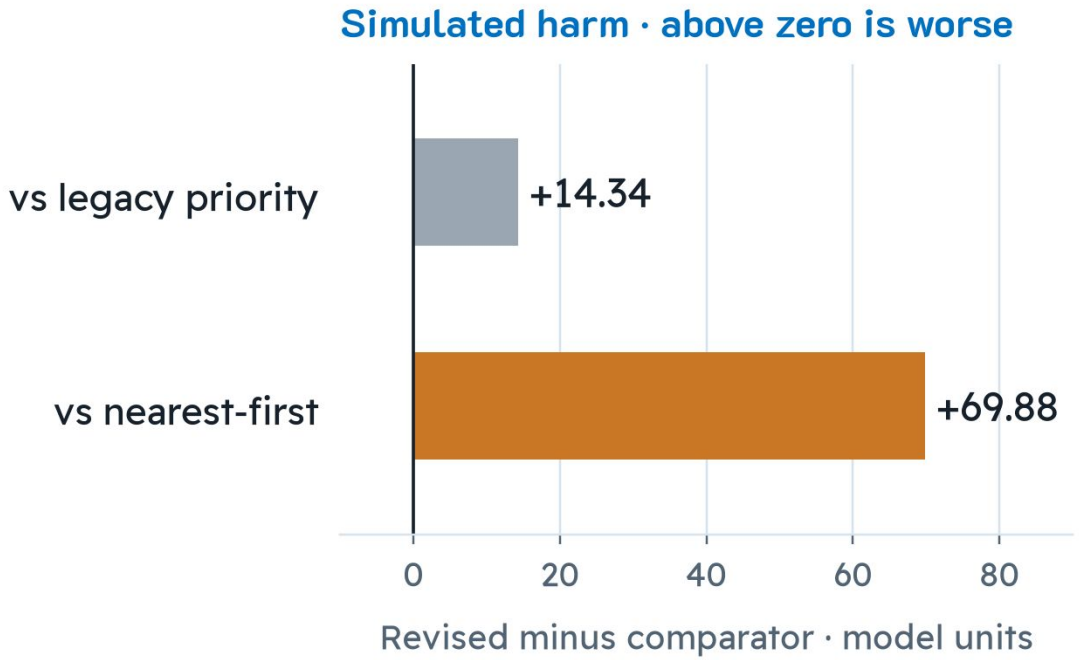
Zero drift at 2×, 5×, 10×
(fixed families and inputs)

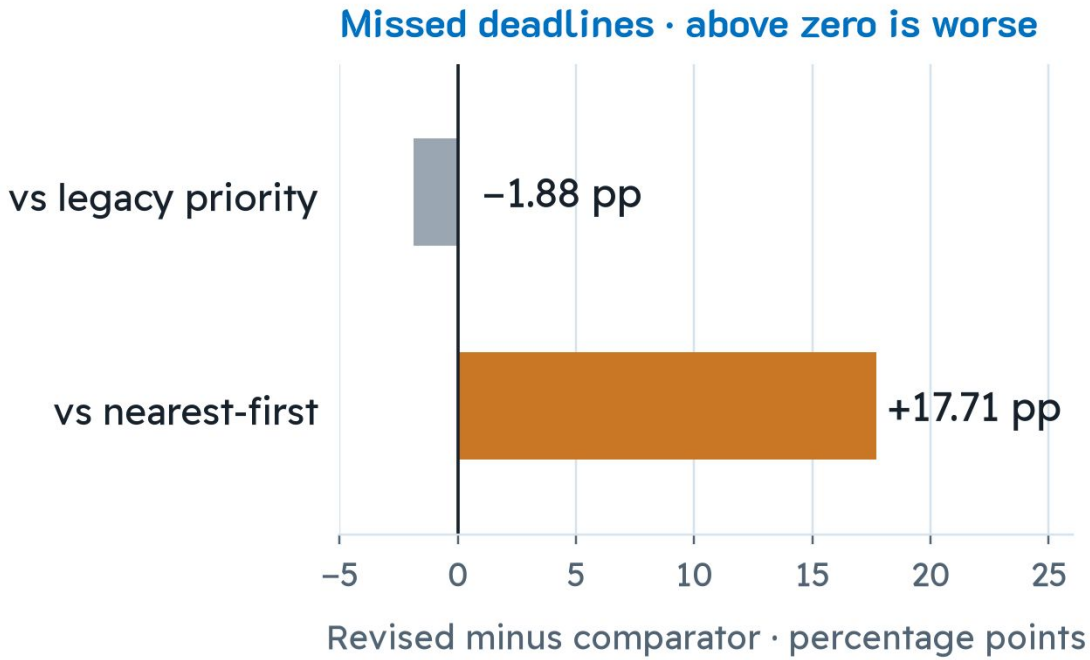
Still vulnerable

Coordinated campaign: drift
.1688 vs .0309 for legacy

Results: Rescue planning

Revised priority minus comparator · Product partition · 40 seed pairs · Holm-adjusted p





Versus legacy priority

Harm p = .563 · deadlines p = .247

Neither difference is significant

Versus nearest-first

Harm and deadlines: p < .001

Both differences are significant

Grouping errors reach the plan

Product vs matched-density Additive:

harm +18.34 · Holm p = .000168

Conclusion & Future work: What comes next

Conditional safeguards are useful properties, not evidence of field effectiveness

Local guarantees do not establish better rescue decisions.

Clustering

Spatial gating ≠ proven clustering gain

Ranking

Copy stability ≠ benefit alignment

Dispatch

A bounded score ≠ a better schedule

Contribution	Controlled tests of each step, with truth held by the evaluator · public artifact [12]
Next	Independently labeled real reports · expert and source checks · joint ranking and scheduling

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Methods and evaluation

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Data and application context

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https://github.com/ngthtrong/nckh2/releases/tag/v1.0.1



THANK YOU

Questions and discussion

Clustering safeguards, ranking stability and dispatch benefit require separate evaluation.

Paper 6444 · Can Tho University

Research artifact [12]: github.com/ngthtrong/nckh2/releases/tag/v1.0.1

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Backup: Evaluation details

Scope and protocol details kept outside the main talk

Separate configurations	Generator 3.0: geographic clustering stress. Candidate 4.1: synthetic ranking / dispatch.
Information boundary	RQ2 uses oracle incident groups. RQ3 schedules predicted clusters; truth is evaluator-only.
Inference unit	40 test seed pairs; three resource conditions are averaged within each seed.
Guarantee conditions	Exact copies preserve scores only with fixed evidence families and fixed score inputs.

Backup: Anticipated questions

Statistics and design choices behind the main results

Interval vs. test (RQ1)

Product − Additive = −0.0118, unadjusted bootstrap CI [−0.0226, −0.0024]; Wilcoxon raw p = .0987, Holm p = .2560. Different procedures; no advantage is claimed.

Louvain vs. Leiden

Product Leiden ARI .9165 vs. Product Louvain .9072; Holm p = .1120.

Fixed constants

21 one-at-a-time settings on 40 test seeds: every mean NDCG@5 span is below .01 (largest .00546).

Why synthetic data

Known incident truth and controlled perturbations allow failure diagnosis. No field validity is claimed.